Integration — Lesson 6 Test: The Definite Integral, the Official Symbol

10 questions · show your work in the box · calculators allowed · draw a quick sketch whenever it helps

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Rain falls at a rate of r(t) millimeters per hour. Write the definite integral that means "the total rain that fell from hour 1 to hour 5." 1 pt

2. In the integral $\int_0^8 (3x + 2)\,dx$, identify: (a) the integrand, (b) the lower limit, (c) the upper limit, and (d) what the dx stands for. 1 pt

3. Evaluate $\int_1^5 6\,dx$ using geometry. Name the shape you used. 1 pt

4. Evaluate $\int_0^3 4x\,dx$ using geometry. Name the shape you used. 1 pt

5. Evaluate $\int_5^5 (x^3 + 9)\,dx$, and explain in one sentence why no computing is needed. 1 pt

6. Evaluate $\int_0^4 (x + 2)\,dx$ using geometry. (Hint: find the height of the graph at each end first.) 1 pt

7. A car's speed after t hours is v(t) miles per hour. Translate $\int_2^6 v(t)\,dt$ into a plain-English sentence, including the units of the answer. 1 pt

8. You are told $\int_0^3 f(x)\,dx = 8$ and $\int_3^9 f(x)\,dx = 11$. Find $\int_0^9 f(x)\,dx$, and name the property you used. 1 pt

9. You are told $\int_0^{12} g(x)\,dx = 40$ and $\int_0^5 g(x)\,dx = 15$. Find $\int_5^{12} g(x)\,dx$. 1 pt

10. A faucet's flow rate f(x) (liters per minute, x in minutes) follows this graph: it climbs along a straight line from 0 (at x = 0) up to 6 (at x = 3), then holds steady at 6 from x = 3 to x = 5. Evaluate $\int_0^5 f(x)\,dx$ and say what the number means, with units. 1 pt